

sensors, and more sophisticated software stacks.

What are the benefits of hardware sensor fusion compared to software-based sensor fusion on a CPU?

We designed this chip for devices such as safety pendants that connect multiple sensors, including sound, motion, pressure, temperature, and vibration, allowing AI to process complex tasks efficiently. Running this workload on a CPU would rapidly drain the battery, so we developed a hardware-based Sense Mesh that fuses sensor data with minimal power consumption.

The chip can operate on a small battery for up to six months at 100 microwatts, whereas traditional software-based approaches can consume 10x more power. The Sense Mesh feeds data directly to always-active AI cores that continue processing even in subconscious mode, supporting up to 20 digital and eight analogue sensors. This architecture enables practical, always-on AI in battery-powered devices.

How do you design an efficient data pipeline so that multiple sensor inputs do not become a bottleneck?

The answer is sustained engineering effort and creative problem-solving. Our AI architecture operates at just 100 microwatts, but existing digital microphones and ADCs were too power-hungry.

We therefore designed our own 5-microwatt ADC, once considered impractical, which now allows the Sense Mesh to process multiple sensor inputs efficiently without bottlenecks.

Can you discuss the new innovations you are currently working on?

Our scalable technology supports devices ranging from 10-core ultra-low-power implementations to a 64-core launch planned for 2026-27, delivering high compute performance with minimal energy consumption. We are also developing devices with thousands of cores, far exceeding current efficiency benchmarks.

The primary challenge remains the long development cycle before revenue generation, which many investors find difficult to support.

Do you think transistor scaling limitations will impact the future of AI-native processors?

Over the past 20 years, innovation in materials, transistors, and EDA tools has slowed. While smaller process nodes have improved density, fundamental progress has stalled, creating bottlenecks in multi-layer AI, memory, and power delivery.

Emerging technologies such as graphene and quantum computing remain distant, making solutions like ours essential for the next 15-20 years. **EFY**

Practical Thinking: Smart Energy Meter-Equipped with Diotec MOSFET DIJ006N90

In smart energy meters, a reliable and efficient power supply is essential for continuous operation. This is achieved using a Switched-Mode Power Supply (SMPS), where a high-voltage MOSFET serves as the main switching element. The MOSFET rapidly turns on and off to convert high-voltage DC (from rectified AC mains) into regulated low-voltage DC needed by the meter microcontroller and communication modules. Its fast switching, high voltage tolerance and low power loss are critical for the SMPS to operate efficiently and safely in harsh grid environments. The MOSFET performance directly impacts the meter overall reliability, energy efficiency and long-term stability in the field.



In smart energy meter power supply, the high-voltage MOSFET acts as a main switching element. Low RDS(on) plays a crucial role while enhancing system, to meet these condition, the DIJ006N90 from Diotec Semiconductor is an 6 A/900 V N-channel MOSFET ready to deliver highest Voltage capability is up to 900V, requiring less Series connection. The very low on-state resistance of typically 2.1 Ohm helps improve the heat dissipation performance of MOSFET, reduce the volume and improve the power density of MOSFET.

FOR MORE INFO



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